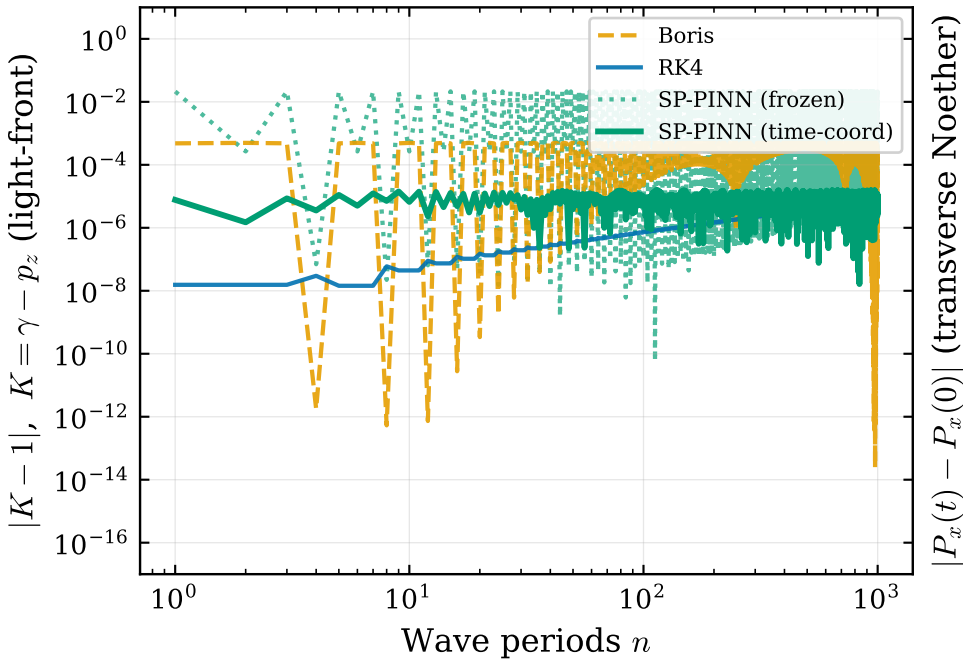


(a) time-related invariant



(b) transverse canonical momentum

